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THE AI COMPANY

Diagnostic Error Code Manual
Knowledge Base
V 0.1

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1. Diagnostic Trouble Code

P2463-00

1. DTC Overview

Parameter	Description
DTC	P2463-00
DTC Description	DPF - Soot load - Stage 2 (> 8.5 g/Ll)
Possible Causes	• Excessive soot load in DPF • Primary fault codes • Blocked differential pressure sensor tubes • Leak in differential pressure sensor tubes • Exhaust gas leak • Excessive ash in DPF • Engine is producing more soot
System Category	Exhaust After-Treatment / DPF Management
Primary Impact	Continued operation risks DPF thermal runaway, substrate cracking, or permanent blockage requiring replacement. Regeneration cycles may become ineffective.

IMPORTANT — PRIMARY ERROR CODES TO CHECK FIRST Before diagnosing the DTC-P2463-00, check and resolve all Primary Error Codes listed below. If any of these codes are active, soot load estimation will be incorrect or soot load will be artificially elevated — making DTC diagnosis unreliable.

2. Primary Error Codes

Before diagnosing the DTC - P2463-00, check and resolve all Primary Error Codes listed below. The Primary Error Codes can exist in two categories:

1. **Estimation Wrong**
2. **Soot Load Increased**

Check for the "Active Error Codes" in both the above categories & resolve them as per appropriate error code troubleshooting steps

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Category 1 — Estimation Wrong

DTC	Fault Code	Description
P245E-11	**FC1881**	DPF differential pressure sensor signal voltage < 0.249 V
P2452-12	**FC3133**	DPF differential pressure sensor signal voltage > 4.75 V
P245E-1F	**FC1883**	DPF differential pressure value > 1.5 kPa at Key ON or Key OFF

Category 2 — Soot Load Increased

DTC	Fault Code	Description
P1194-00	**FC1981**	DPF differential pressure sensor in range but value too high (> 9 g/L)
P2459-00	**FC3375**	Frequent Regeneration
P24A2-00	**FC3376**	Combined soot load not reduced after complete Active Regen
P1629-00	**FC2777**	Regen inhibit switch ON for extended duration

Additional Checks Required

- Check any DPF temperature sensor module-related error codes in active condition.
- Check any HC metering unit-related error codes in active condition.

2. Diagnostic Procedure — DTC P2463-00

2.1 Primary Error Codes – Estimation Wrong

1. P245E-11 / FC1881

Fault Description	DPF differential pressure sensor signal voltage is less than 0.249 V

Fault Type	Estimation Wrong — sensor signal out of range (low)
Impact	ECU receives incorrect differential pressure data → Soot load calculation unreliable for P2452

Step 1 — Connect the Diagnostic Scan Tool and Confirm Active Fault Code

1. Turn the ignition switch to the OFF position.
2. Locate the **OBD-II Diagnostic Port**, as per the model of the vehicle
3. Firmly plug the diagnostic scan tool connector into the OBD-II port.
4. Turn the ignition key to the ON position (engine must remain OFF).
5. Power on the diagnostic scan tool.
6. Open the Diagnostic App in the Tab & establish connection to the vehicle.
7. Once the connection is established between the VCI and the Diagnostic App, Start the Diagnosis

Expected Result	Scan tool displays stored fault codes including P245E-11 / FC1881 as active or inactive
Next Action	If the Primary Error Code is active, proceed with the Step-2. If not, ignore & proceed the diagnosis with the next Primary Error Codes

Step 2 — If P245E-11 / FC1881 is Active

Record Freeze Frame / Snapshot Data

1. On the scan tool, navigate to Freeze Frame Data or Snapshot Data for fault P245E-11.
2. Record the following parameters at the time the fault was triggered:
   - DPF Differential Pressure Sensor Voltage (target: 0.249 V – 4.75 V)
   - Engine RPM
   - Coolant Temperature
   - Vehicle Speed
   - DPF upstream and downstream temperature readings

3. Save or photograph the freeze frame for reference during further diagnosis.

Expected Result	Freeze frame data should show sensor voltage reading below 0.249 V confirming the fault trigger condition.

Step 3 — Perform Visual Inspection of the DPF Differential Pressure Sensor & Wiring

1. Turn the ignition to OFF and wait 2 minutes for the system to de-energize.
2. The Diesel Particulate Filter (DPF) assembly is part of the vehicle's exhaust after-treatment system. It is installed along the exhaust pipe and is responsible for trapping soot particles produced during diesel combustion.
3. Where the DPF Assembly Is Located?
   - The DPF assembly is typically installed in the exhaust line underneath the vehicle.
   - It is positioned:
   - Downstream Of the Turbocharger and Diesel Oxidation Catalyst (DOC)
   - Before The Muffler and Tailpipe
   - Usually Below the Engine or Slightly Toward the Middle of the Vehicle Chassis
   - When Looking Under the Vehicle, The DPF Assembly Appears as A Large Metal Cylindrical or Box-Shaped Chamber Integrated into The Exhaust Pipe. It Is Larger Than a Normal Exhaust Pipe Section and May Have Temperature Sensors, Pressure Ports, And Wiring Connectors Attached to It.
4. Inspect both pressure hoses connected to the sensor:
   - Check for cracks, splits or collapsed sections
   - Ensure hoses are firmly seated on their fittings — no looseness or signs of blowout
   - Check for soot or moisture deposits inside hose ends (indicates leaks)
5. Inspect the sensor electrical connector:
   1. Check for pushed-back, bent, or corroded pins
   2. Check for water ingress or mud inside the connector housing
   3. Confirm the connector locking tab is fully engaged (audible click)
6. Inspect the sensor body:

- Look for physical damage, cracks, or impact marks
- Verify the sensor is mounted securely (no looseness)

7. Inspect the wiring harness from the sensor back to the ECU connector:
   - Look for chafing against sharp edges, exhaust heat shields, or moving parts
   - Check for any pinched or crushed sections

Expected Result	No visible damage, disconnections, hose leaks, or wiring faults found.
If Fault Found	Repair damaged wiring, replace connector, or reseat loose hoses as required before continuing.

Step 4 — Check Sensor Supply Voltage

WARNING: Ensure the area around the DPF and exhaust is cool before working near the sensor. Risk of burns.

1. Set the Digital Multimeter (DMM) to DC Voltage mode.
2. Turn the ignition to the ON position (engine OFF).
3. Locate the sensor connector — DO NOT disconnect it for this test. Use a back-probe or a breakout harness.
4. Identify sensor connector pin functions (refer to the vehicle-specific wiring diagram):
   - Pin A: Signal wire (0.249 V – 4.75 V nominal range)
   - Pin B: Sensor ground
   - Pin C: 5V reference supply from ECU
5. Probe Pin C (5V reference) against a known chassis ground.
6. Record the voltage reading.

Expected Result	Voltage should read 4.75 V – 5.25 V (5V ± 5%). If reading is 0 V or less than 4.5 V, proceed to Step 5.
If Voltage Absent or Low	Diagnose the ECU 5V reference supply circuit — check fuses, wiring continuity, and ECU connector pins before replacing the sensor.

Step 5 — Check Sensor Ground Circuit

1. With ignition ON, probe Pin B (sensor ground) against battery negative.
2. Record the voltage reading.

Expected Result	Voltage should read 0 V – 0.1 V (near zero). A reading above 0.5 V indicates a high resistance ground fault.
If Ground Fault Found	Inspect the ground path from the sensor connector back to the ECU ground point and vehicle chassis ground. Clean or repair as needed.

Step 6 — Check Sensor Signal Wire Continuity

1. Turn the ignition to OFF.
2. Disconnect the sensor electrical connector.
3. Disconnect the ECU connector at the ECU end (refer to workshop manual for ECU connector location and pin assignment for the DPF pressure sensor signal wire).
4. Set DMM to Resistance (Ω) mode.
5. Probe between the signal pin at the sensor connector and the corresponding pin at the ECU connector.
6. Record the resistance value.

Expected Result	Resistance should be less than 2 Ω. A reading above 2 Ω indicates a break or high-resistance fault in the signal wire.
If Open Circuit	Repair or replace the signal wire between the sensor and ECU. Clear codes and retest after repair.

Step 7 — Check for Short to Ground on Signal Wire

1. With sensor connector and ECU connector still disconnected (from Step 6).
2. Set DMM to Resistance (Ω) mode.
3. Probe between the signal pin at the sensor connector and chassis ground.
4. Record the resistance value.

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Expected Result	Resistance should read OL (over-limit / infinite). A low resistance reading (below 10 kΩ) confirms a short to ground on the signal wire — which would pull signal voltage below 0.249 V and set P245E-11.
If Short to Ground Found	Locate the point of insulation damage or contact with the vehicle body/exhaust and repair. Clear codes and retest.

Step 8 — Replace DPF Differential Pressure Sensor (if wiring is confirmed good)

1. Ensure the ignition is OFF and the exhaust system is fully cooled.
2. Label and disconnect both pressure hoses from the faulty sensor.
3. Disconnect the sensor electrical connector.
4. Remove the sensor mounting fastener(s) and remove the sensor.
5. Install the new OEM DPF differential pressure sensor — do not use aftermarket substitutes for emissions-critical sensors.
6. Reconnect and firmly seat both pressure hoses.
7. Reconnect the electrical connector (confirm locking tab clicks).
8. Turn the ignition ON and check scan tool live data to confirm sensor signal voltage is within 0.249 V – 4.75 V range.

Expected Result	Sensor signal voltage reads within specification. No active fault codes for P245E-11.

Step 9 — Clear Fault Codes and Verify Repair

1. On the scan tool, navigate to: Clear / Erase Diagnostic Trouble Codes.
2. Clear all stored DTCs.
3. Re-read DTCs after the test cycle & ensure Error Code **P245E-11 / FC1881 is absent**

Expected Result	P245E-11 / FC1881 does not return. If the fault returns, re-inspect all wiring and connections or escalate to ECU diagnostics.

2. P2452-12 / FC3133

Fault Description	DPF differential pressure sensor signal voltage is more than 4.75 V
Fault Type	Estimation Wrong — sensor signal out of range (high)
Impact	ECU receives a sensor signal outside the calibrated voltage range (>4.75 V). This prevents accurate exhaust pressure measurement and may interrupt DPF regeneration control.

Step 1 — Connect the Diagnostic Scan Tool & confirm Active Code

1. Turn the ignition switch to the OFF position.
2. Locate the **OBD-II Diagnostic Port**, as per the model of the vehicle
3. Firmly plug the diagnostic scan tool connector into the OBD-II port.
4. Turn the ignition key to the ON position (engine must remain OFF).
5. Power on the diagnostic scan tool.
6. Open the Diagnostic App in the Tab & establish connection to the vehicle.
7. Once the connection is established between the VCI and the Diagnostic App, Start the Diagnosis

Expected Result	Scan tool displays stored fault codes including P2452-12 / FC3133 as active or inactive
Next Action	If the Primary Error Code is active, proceed with the Step-2. If not, ignore & proceed the diagnosis with the next Primary Error Code

Step 2 — Read Live Sensor Data — Confirm Over-Voltage Condition

1. On the scan tool, navigate to: Live Data / Data Stream.
2. Select and monitor the DPF Differential Pressure Sensor Voltage parameter.
3. With ignition ON and engine OFF, observe the voltage reading.
4. Record the live sensor voltage value.

Expected Result	Sensor voltage should be reading above 4.75 V confirming the fault condition. A voltage above 4.75 V typically indicates: short to battery positive on signal wire, internal sensor fault, or a blocked/disconnected pressure hose causing pressure overshoot.

Step 3 — Inspect Pressure Hoses for Blockage or Disconnection

1. Turn the ignition OFF. Allow the exhaust and DPF to cool completely before touching.
2. Locate both pressure hoses on the DPF differential pressure sensor:
3. The pressure hoses for the **DPF Differential Pressure Sensor** are located in the engine compartment and are attached directly to the sensor body through two small hose ports.
4. To find them, open the hood and locate the DPF differential pressure sensor mounted on a bracket near the firewall or close to the exhaust system.
5. From the sensor, two thin rubber or silicone hoses extend downward toward the exhaust system.
6. One hose connects to a pressure tap on the exhaust pipe **before** the **Diesel Particulate Filter** (upstream/high-pressure side), while the other connects to a pressure tap **after** the DPF (downstream/low-pressure side).
7. These hoses allow the sensor to measure the pressure difference across the DPF to determine soot accumulation.
8. Check whether either hose has become disconnected or splits — a disconnected upstream hose or a blocked downstream hose can force the sensor into an over-pressure, over-voltage state.
9. Carefully disconnect each hose and visually check the internal bore for soot blockage.
10. Use a thin wire probe or compressed air (low pressure, max 2 bar) to check for free airflow through each hose.
11. Reconnect all hoses firmly.

Expected Result	Both hoses are free from blockage, properly connected, and undamaged.
If Hose Blockage Found	Clear the blockage with compressed air. If the hose bore is damaged or cannot be cleared, replace the hose. Clear codes and retest.

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Step 4 — Check for Short to Battery (B+) on Signal Wire

1. Turn the ignition OFF.
2. Disconnect the DPF differential pressure sensor electrical connector.
3. Disconnect the relevant ECU connector pin for the sensor signal wire.
4. Set the DMM to DC Voltage mode.
5. Turn the ignition to ON.
6. Probe the signal wire (at the sensor connector side) against chassis ground.
7. Record the voltage.

Expected Result	Voltage should read 0 V with sensor and ECU disconnected. A reading of battery voltage (12 V – 14 V) confirms a hard short to B+ on the signal wire.
If Short to B+ Found	Trace the signal wire harness for contact with battery positive supply wires or a chafed section touching a live circuit. Repair insulation or replace the wire segment. Clear codes and retest.

Step 5 — Replace DPF Differential Pressure Sensor if All Wiring Checks Pass

1. If supply voltage, ground, signal wire, and hose checks are all within specification, the sensor itself has an internal fault causing high voltage output.
2. Turn ignition OFF. Allow exhaust to cool.
3. Disconnect both pressure hoses and the electrical connector from the sensor.
4. Remove and replace the sensor with a new OEM part.
5. Reconnect hoses and connector. Confirm no air leaks at hose connections.
6. Turn ignition ON. Verify live sensor voltage is now within 0.249 V – 4.75 V.

Expected Result	Sensor voltage reads within specification. P2452-12 / FC3133 no longer active.

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Step 6 — Clear Fault Codes and Verify Repair

1. Clear all DTCs via the scan tool.
2. Perform the required DPF diagnostic drive cycle.
3. Re-read DTCs after the test cycle & ensure Error Code **P2452-12 / FC3133 is absent**

Expected Result	P2452-12 / FC3133 does not re-trigger. Proceed to diagnose remaining pre-check codes.

3. P245E-1F / FC1883

Fault Description	DPF differential pressure value is greater than 1.5 kPa at Key ON or Key OFF condition (engine not running)
Fault Type	Estimation Wrong — pressure sensor does not read zero (atmospheric reference) at rest
Impact	ECU baseline reference is offset → all subsequent soot load calculations skewed → P2452 diagnosis unreliable

Client Note: *The 1.5 kPa threshold is based on the client-provided diagnostic specification.*

Step 1 — Connect the Diagnostic Scan Tool and Confirm Active Fault Code

1. Turn the ignition switch to the OFF position.
2. Locate the **OBD-II Diagnostic Port**, as per the model of the vehicle
3. Firmly plug the diagnostic scan tool connector into the OBD-II port.
4. Turn the ignition key to the ON position (engine must remain OFF).
5. Power on the diagnostic scan tool.
6. Open the Diagnostic App in the Tab & establish connection to the vehicle.
7. Once the connection is established between the VCI and the Diagnostic App, Start the Diagnosis

Expected Result	Scan tool displays stored fault codes including P245E-1F / FC1883 as active or inactive
Next Action	If the Primary Error Code is active, proceed with the Step-2. If not, ignore & proceed the diagnosis with the next Primary Error Code

If the value is greater than **1.5 kPa**, the fault condition described by the client specification is confirmed.

Step 2 — Inspect Pressure Hoses Connected to the Sensor

1. Turn the ignition OFF and allow the exhaust system to cool.
2. Locate the DPF Differential Pressure Sensor in the engine compartment.
3. Identify the two pressure hoses attached to the sensor ports.
4. Inspect both hoses for:
   - soot buildup or blockage
   - cracks or physical damage
   - loose or disconnected hose fittings
   - kinks restricting airflow
5. If necessary, remove the hoses and inspect them internally for soot accumulation.

Expected Result	Pressure reading drops to below 1.5 kPa at key ON after clearing hose blockages.
If Blockage Was Found and Cleared	Clear fault codes, retest, and verify P245E-1F does not return.

Step 3 — Inspect Sensor Wiring

1. Inspect the electrical connector attached to the DPF differential pressure sensor.
2. Check the wiring harness for:
   - damaged insulation
   - loose connectors
   - corroded terminals
   - broken wires

Expected Result	Wiring and connectors should be intact and securely connected.

Step 4 — Replace the DPF Differential Pressure Sensor (If Required)

If hoses and wiring are functioning correctly but the fault persists, the sensor itself may be faulty.

1. Turn ignition OFF and allow the exhaust system to cool.
2. Disconnect the electrical connector from the sensor.
3. Remove the pressure hoses connected to the sensor.
4. Remove the sensor from its mounting bracket.
5. Install a new OEM DPF differential pressure sensor.
6. Reconnect hoses and the electrical connector securely.

Expected Result	After replacement, the pressure reading should return to a normal value when the engine is not running.

Step 5 — Clear Fault Codes and Verify

1. Use the scan tool to clear stored diagnostic trouble codes.
2. Turn the ignition OFF and then ON again.
3. Re-read DTCs after the test cycle & ensure Error Code **P245E-1F / FC1883 is absent**

Expected Result	The diagnostic trouble code should not reappear if the issue has been resolved.

i. Items That May Require Client/OEM Clarification

The following procedures were not confirmed in the referenced public source and may require additional OEM documentation:

- Sensor zero-point calibration or offset reset
- Exact DPF regeneration verification drive cycle
- Manufacturer-specific scan tool functions

2.2 Primary Error Codes – Soot Load Increased

1. P1194 /FC1981

Parameter	Description
Fault Description	**P1194 /FC1981-** DPF differential pressure sensor signal in range but value is too high. (>9 gpl)
Fault Type	Soot Load Increased - Engine management / OBDII EATS System

Step 1 — Connect the Diagnostic Scan Tool and Confirm Active Fault Code

1. Turn the ignition switch to the OFF position.
2. Locate the **OBD-II Diagnostic Port**, as per the model of the vehicle
3. Firmly plug the diagnostic scan tool connector into the OBD-II port.
4. Turn the ignition key to the ON position (engine must remain OFF).
5. Power on the diagnostic scan tool.
6. Open the Diagnostic App in the Tab & establish connection to the vehicle.
7. Once the connection is established between the VCI and the Diagnostic App, Start the Diagnosis

Expected Result	Scan tool displays the stored fault code P1194 /FC1981, as active or inactive
Next Action	If the Primary Error Code is active, proceed with the Step-2. If not, ignore & proceed the diagnosis with the next Primary Error Code

Step 2 — Checking for the additional Primary Codes along with P1194 /FC1981

1. Check also if the Primary Fault Codes: P2459-00/FC 3375, P24A2-00/FC 3376 & P1629-00/FC2777 are showing active

Expected Result	The Primary Fault Codes - P2459-00/FC 3375, P24A2-00/FC 3376 & P1629-00/FC2777 are all showing active status.
Next Action	Proceed the diagnosis as per Step-3

Step 3 — Checking the Combined Soot Load value

2. Navigate to the "sensor live data" parameter in the Diagnostic Application
3. Select the "Combined soot load estimate" parameter & check the value

Expected Result	The Combined soot load estimate is showing > 8.5 g/L
Next Action	Proceed the diagnosis as per Step-4

Step 4 — Checking the Rail pressure Sensor Data

1. Under ECU Live data, select Fuel & Check the Rail Pressure Sensor data
2. Check any DPF temperature sensor module related error codes in active condition.
3. Check any HC metering unit related error codes in active condition.

Expected Result	The Raile Pressure Sensor Data showing - 414- 420 bar
Next Action	Proceed the diagnosis as per Step-5

Step 5 — Check pre-conditions for performing DPF Restriction Test

1. Click on the "Diagnostic Scan" tab in the Diagnostic Application.
2. Navigate to the "Actuator Test" under the Engine – Cummins – H6 -N4- BSVI.
3. Select "DPF Choking test"
4. Under the "Precondition Tab, check & confirm all the pre-conditions are met.
5. Refer the SOP below to perform the DPF Restriction Test

DPF Restriction Test SOP:

Purpose:

This test has to be done, if below error codes are found ACTIVE in the system:

- P2459-00 – Frequent Regeneration
- P24A2-00 – Combined soot load not reducing after completion of active regeneration
- P1629-00/FC2777 - Regen inhibit switch ON for longer Duration.

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To identify whether DPF requires cleaning.

- If combined soot load is around 4.5 gpl and more, even after completion of REGEN. This indicates accumulation of ash. This test enables verifying this.
- During this test, Delta pressure sensor will be monitored. If any issue is there, appropriate error code will be triggered in the system

What happens during the test?

- Engine RPM will be raised to 2000 rpm.
- Delta pressure values will be monitored.
- Test will run for 3-4 minutes.
- At the end of the test, engine will come back to idle rpm & result; Pass/Fail will be displayed.

Pre-conditions:

- Restriction test will be initiated within ONE HOUR of completing stationary REGEN.
- Active error codes that have RPM limitations & Delta P sensor/Temperature sensor related should not present in the system.
- Engine operating status should not be in REGEN mode.
- Engine coolant temperature should be above 60 degree centigrade.
- Clutch switch not pressed.
- Brake switch not active
- Parking brake switch status should be active.
- PTO status switch NOT active.
- Accelerator pedal NOT pressed.
- Gear should be in NEUTRAL.
- Engine running in IDLE.

SOP:

- Ensure pre- conditions are met.
- Go to vehicle>> ACU>>Cummins>>Actuators>>AFT DPF restriction test (DPF choking test)
- Select “Restriction Test” before replacement of DPF or after.
- Double click on “STOP” and press the drop down & select “START”.
- Observe the below:
- Engine RPM raises to 2000 rpm.
- This test will run for 3-4 minutes and will stop. Status will indicate the result.

- The test will abort if the accelerator pedal or service brake is depressed at any time.

Test status/Result:

- If any of the pre-conditions are not met either at the time of initiating the test or during the test, those deviations will be displayed in Connectall and test will abort.
- Test status "PASS/FAIL" will be displayed.
- If FAIL: Replace DPF/Clean the DPF only in the AL recommended cleaning centres. In case any new error occurs related to Delta/ Temperature sensor/RPM limitation errors, rectify and do the actuator test again before replacing DPF.
- IF PASS: No issue with DPF. Check as per DSM for "P249F – frequent regeneration error."

Step 6 — Performing DPF Restriction Test

1. As per the SOP, ensure & adhere to check all the pre- conditions
2. Perform the DPF Restriction Test and ensure the test has passed.
3. Re-read DTCs after the test cycle & ensure Error Codes P1194 /FC1981, P2459-00/FC 3375, P24A2-00/FC 3376 & P1629-00/FC2777 are absent

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